Assignment No. 4

Title of the Experiment:
Design and Develop Custom Application (Mini Project) using Salesforce Cloud

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Objectives:

- The objective of this experiment was to design and develop a custom mini project application a Student Management System on the Salesforce Cloud platform. The application demonstrates the creation of custom objects, fields, relationships, formula fields, and a Lightning Experience app, entirely using Salesforce's declarative (point-and-click) development tools without writing code.

Software & Resources Used:

- Platform: Salesforce.com (Developer Edition Org)
- Interface: Salesforce Lightning Experience + Setup > Object Manager
- Custom Objects Created: Student_c, Course_c, Enrollment_c
- App Type: Lightning App (Student Management System)
- Features Used: Custom Objects, Lookup Relationships, Formula Fields, Auto Number, App Manager
- Browser: Google Chrome

Theory / Background:

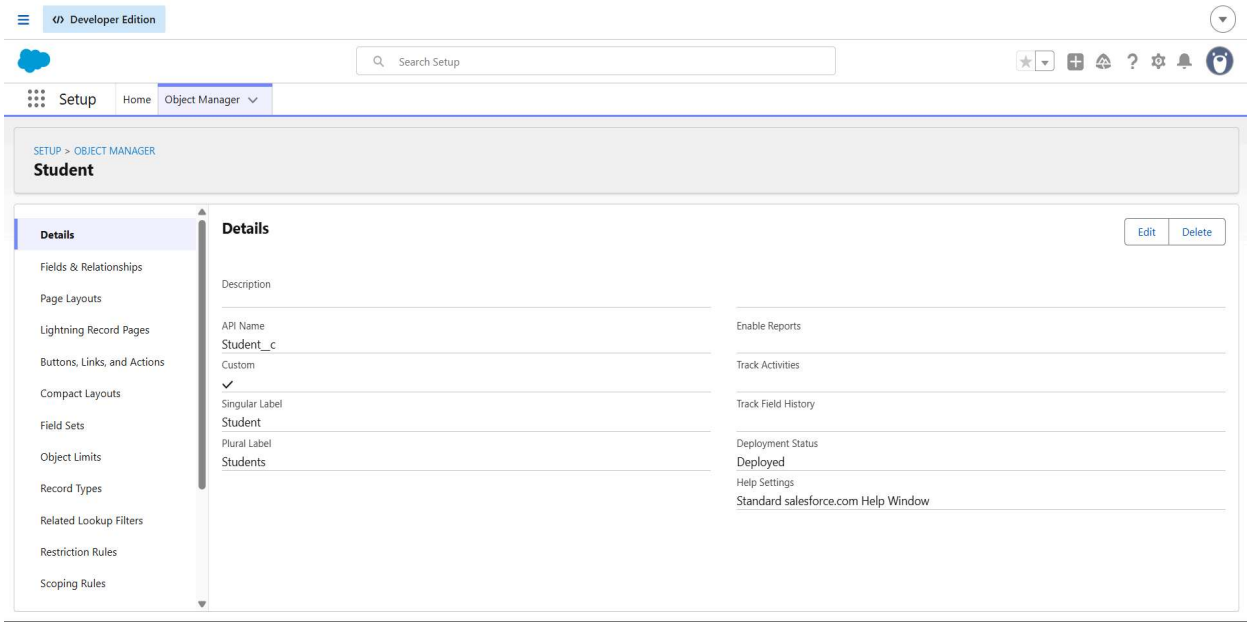
Salesforce is a cloud-based CRM and PaaS platform that allows developers and administrators to build fully custom applications without traditional coding, using its declarative tools. A Custom Object in Salesforce is a user-defined database table that stores data specific to an organization's needs, similar to a table in a relational database.

A Lookup Relationship creates a link between two objects (similar to a foreign key), while a Formula Field is a read-only field whose value is automatically calculated using a formula expression in this case, combining the Student name and Course name into a summary string. The Lightning App Builder and App Manager allow creating custom navigation apps accessible from the Salesforce Lightning UI.

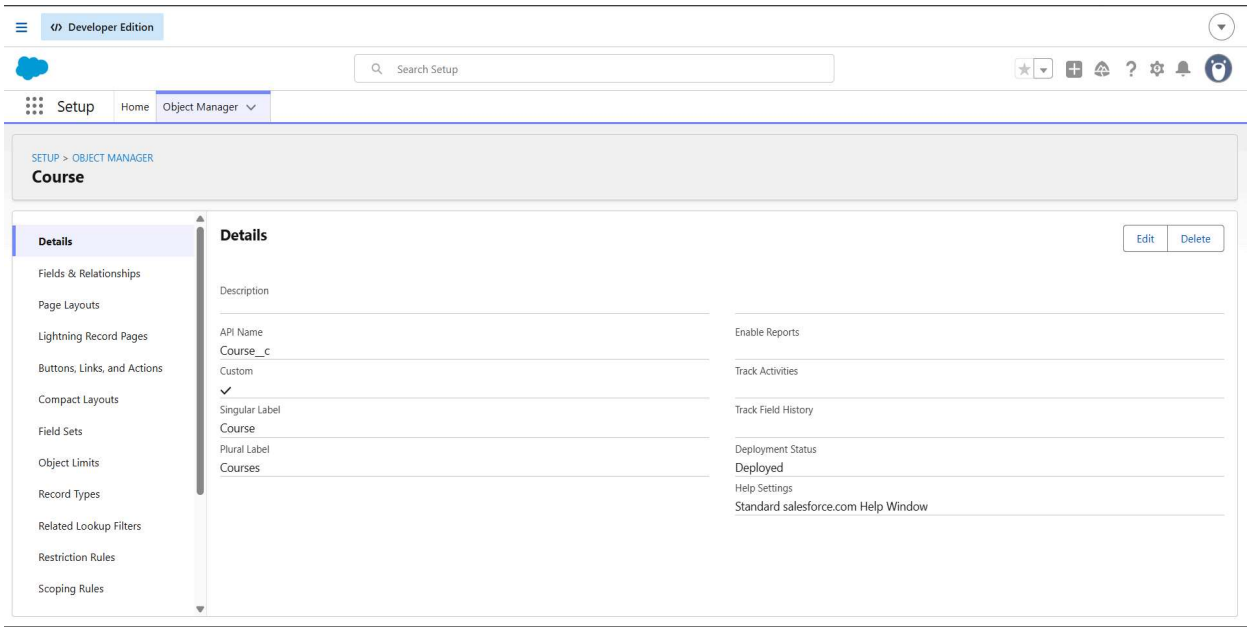
This mini project implements a Student Management System with three objects: Student (stores student details), Course (stores course details), and Enrollment (a junction object linking a Student to a Course, with an auto-generated Enrollment ID and a formula-based Enrollment Summary).

Steps / Procedure with Screenshots:

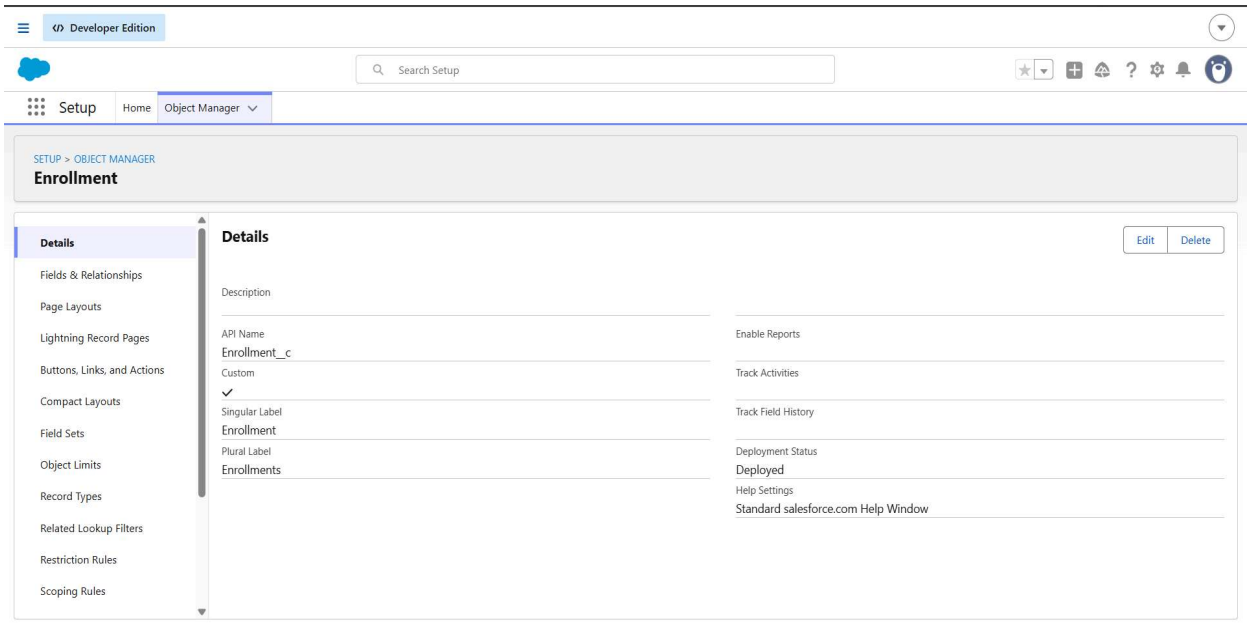
Step 1: We navigated to Setup > Object Manager and created the first custom object 'Student'. The object was configured with API Name Student_c, Singular Label 'Student', Plural Label 'Students', and Deployment Status set to 'Deployed'. The Custom checkbox was checked, confirming it is a user-defined object.



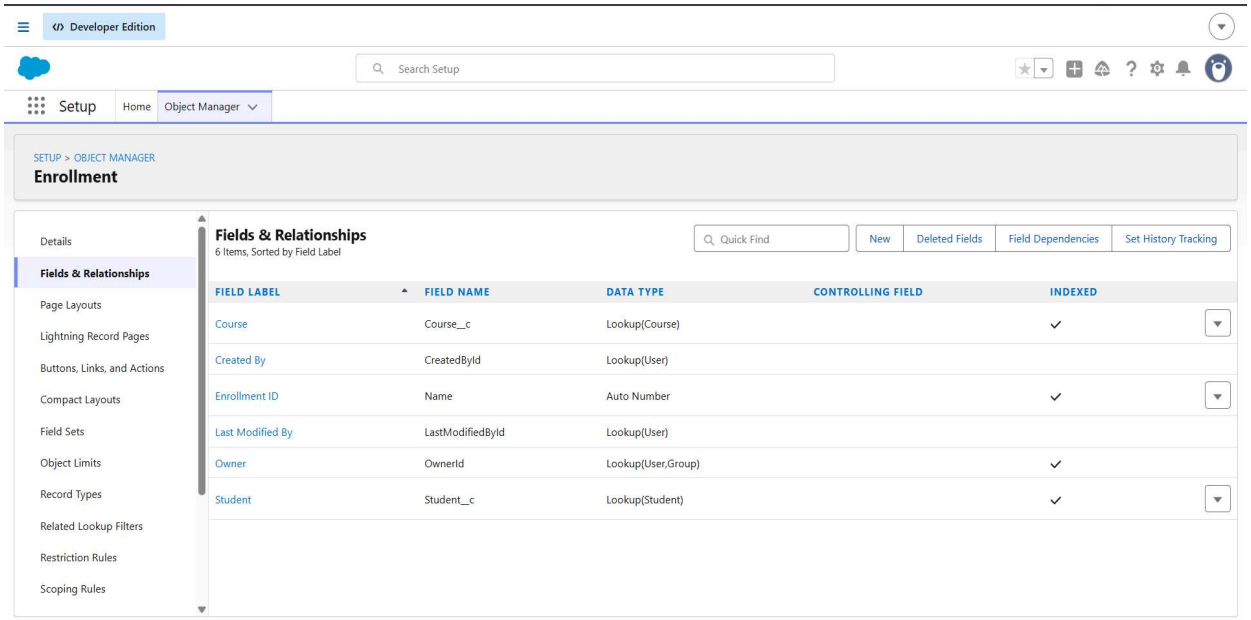
Step 2: We created the second custom object 'Course' with API Name Course_c, Singular Label 'Course', Plural Label 'Courses', and Deployment Status 'Deployed'. This object will store the available courses that students can enroll in.



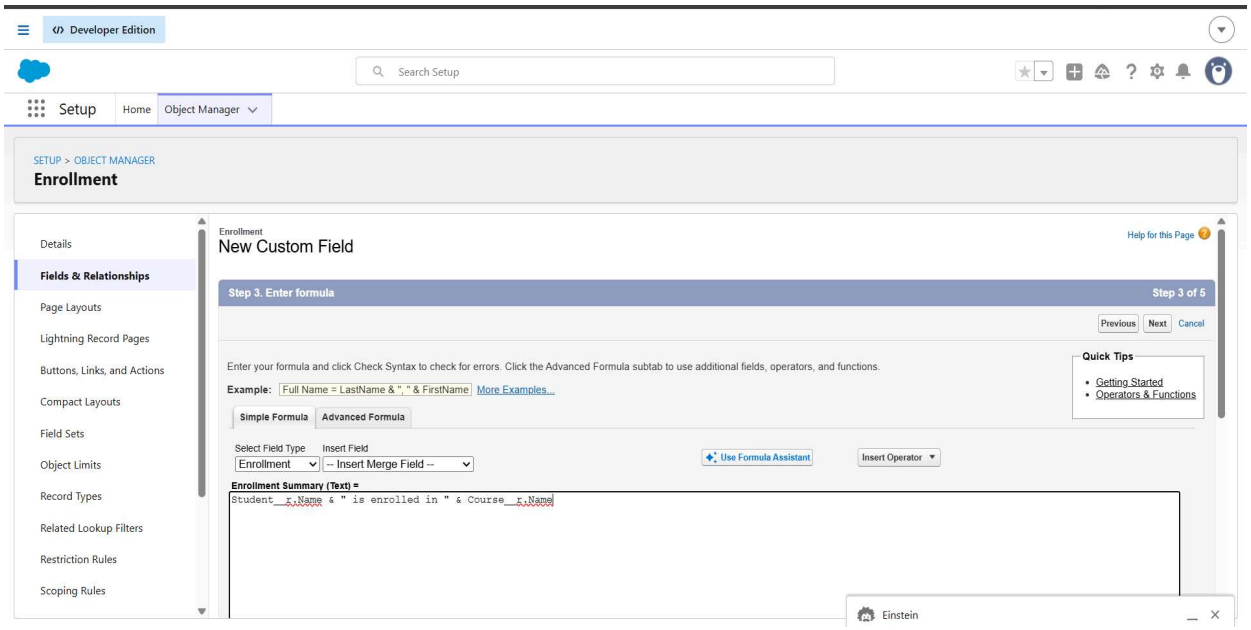
Step 3: We created the third custom object 'Enrollment' with API Name Enrollment_c, Singular Label 'Enrollment', Plural Label 'Enrollments', and Deployment Status 'Deployed'. This object acts as a junction between Student and Course, recording which student is enrolled in which course.



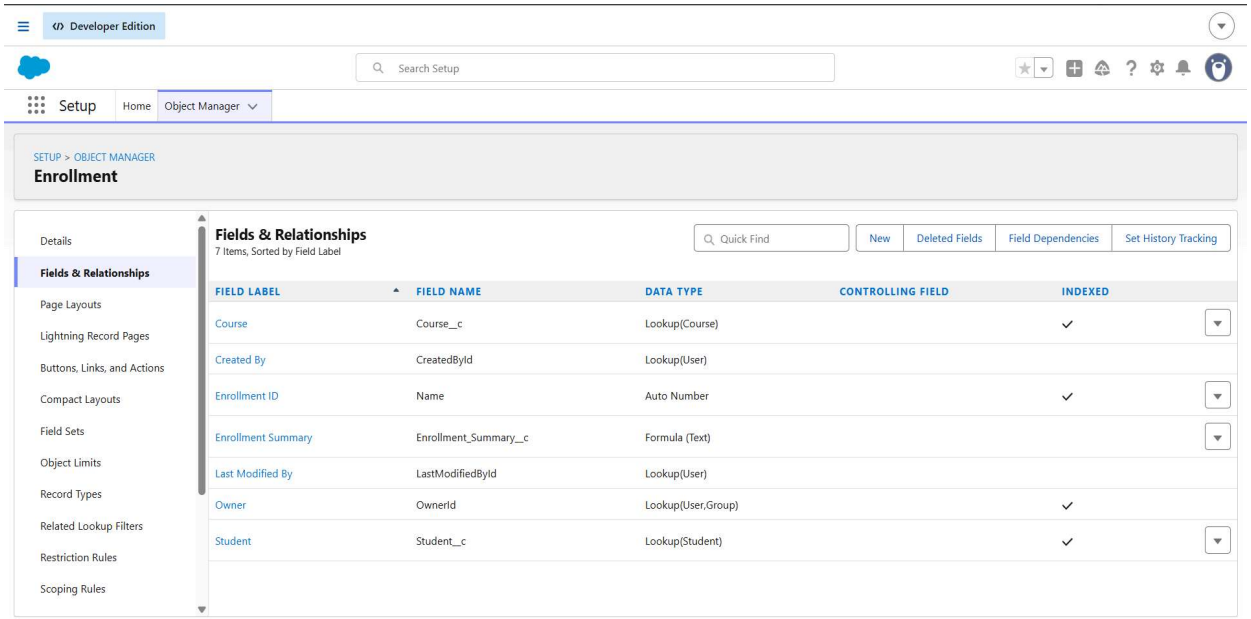
Step 4: We added Fields & Relationships to the Enrollment object. The Fields & Relationships page shows 6 fields: Course_c (Lookup to Course), Enrollment ID (Auto Number - the unique record identifier), Student_c (Lookup to Student), plus system fields CreatedById, LastModifiedById, and OwnerId. The Lookup fields create relational links between Enrollment and the Student and Course objects.



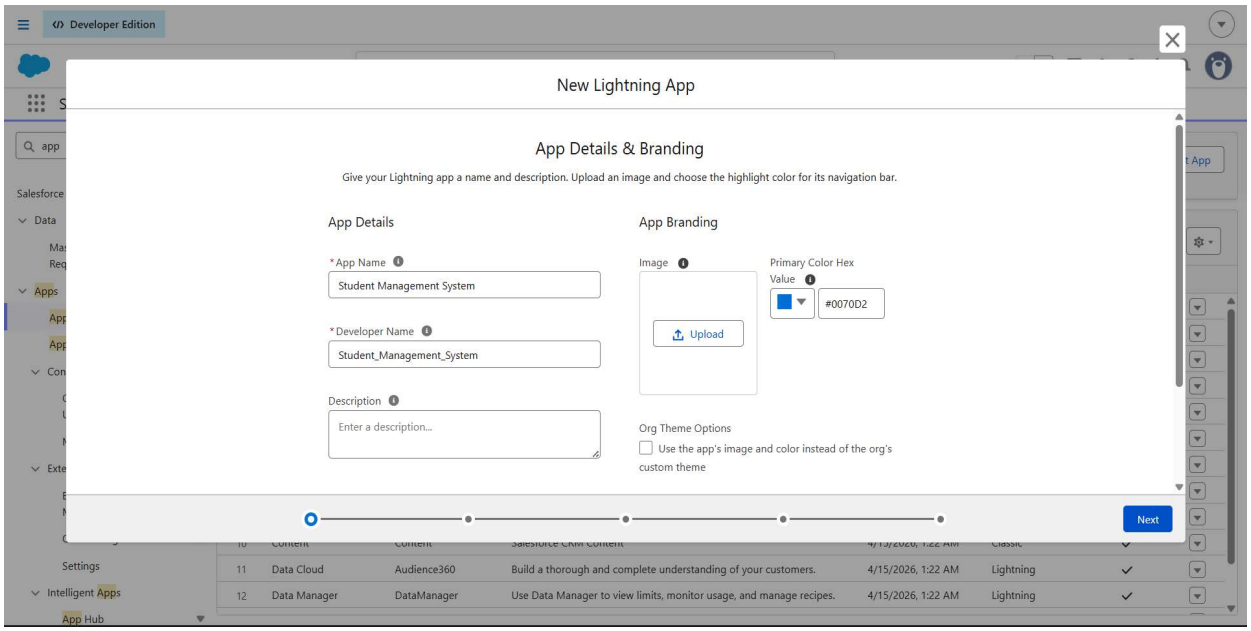
Step 5: We created a custom Formula field named 'Enrollment Summary' on the Enrollment object. In Step 3 of the New Custom Field wizard, we entered the formula: Student_r.Name & " is enrolled in " & Course_r.Name. This formula uses cross-object references to concatenate the related Student's name and Course's name into a readable summary text.



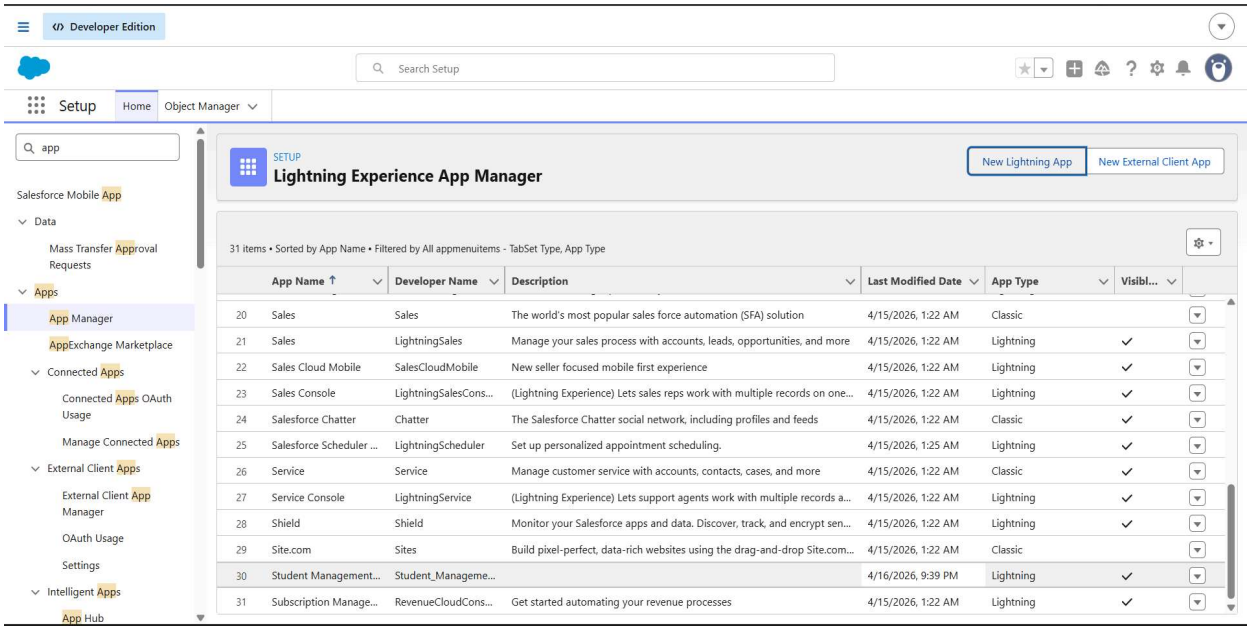
Step 6: After saving the formula field, the Enrollment Fields & Relationships page now shows 7 fields, with the newly added 'Enrollment Summary' (API Name: Enrollment_Summary_c, Data Type: Formula (Text)) visible in the list alongside the Lookup and Auto Number fields.



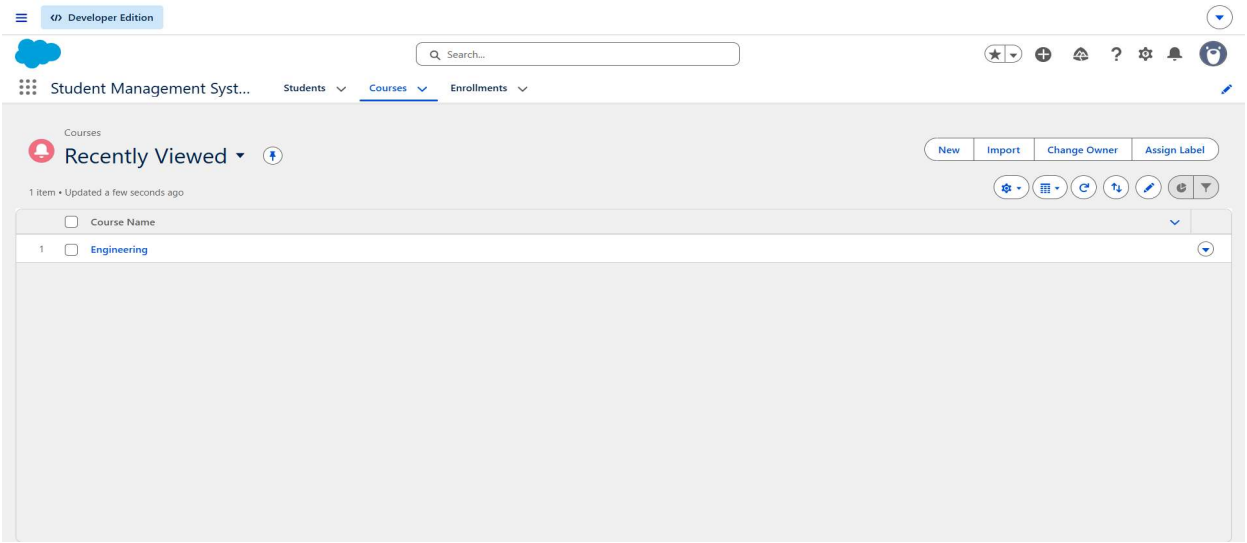
Step 7: We navigated to Setup > Apps > App Manager and clicked 'New Lightning App'. In the App Details & Branding wizard, we entered App Name as 'Student Management System' and Developer Name as 'Student_Management_System', with the primary color set to #0070D2 (Salesforce blue). This creates the navigation app that will expose the Students, Courses, and Enrollments tabs.



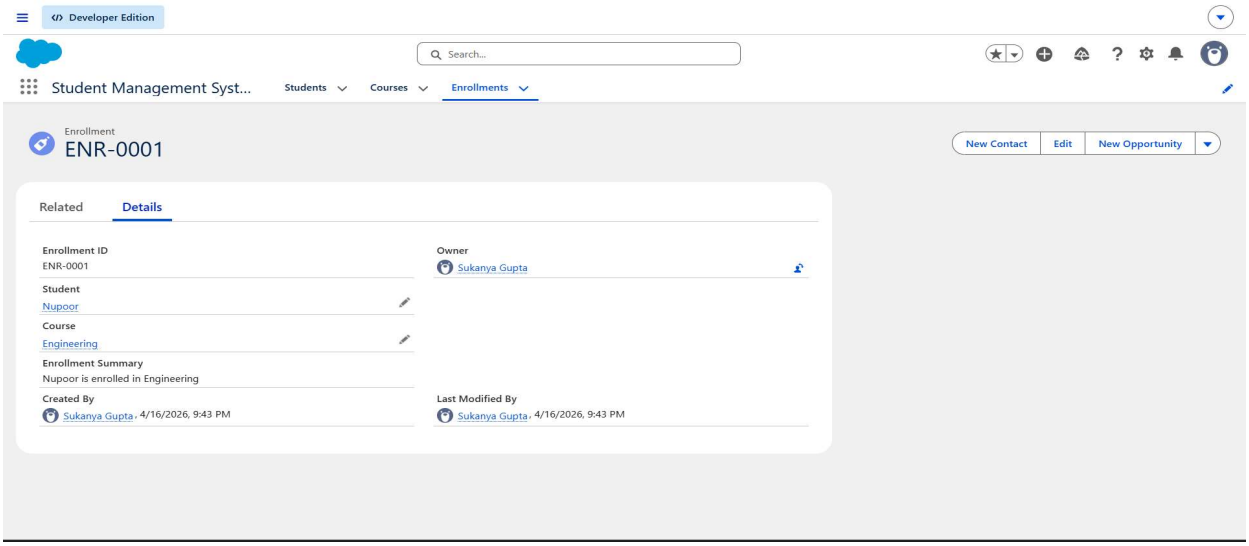
Step 8: The Lightning Experience App Manager confirmed that the 'Student Management...' app (Developer Name: Student_Manageme...) was successfully created as a Lightning app type, visible in row 30 of the App Manager list, with the last modified date 4/16/2026 and the Visible checkbox checked.



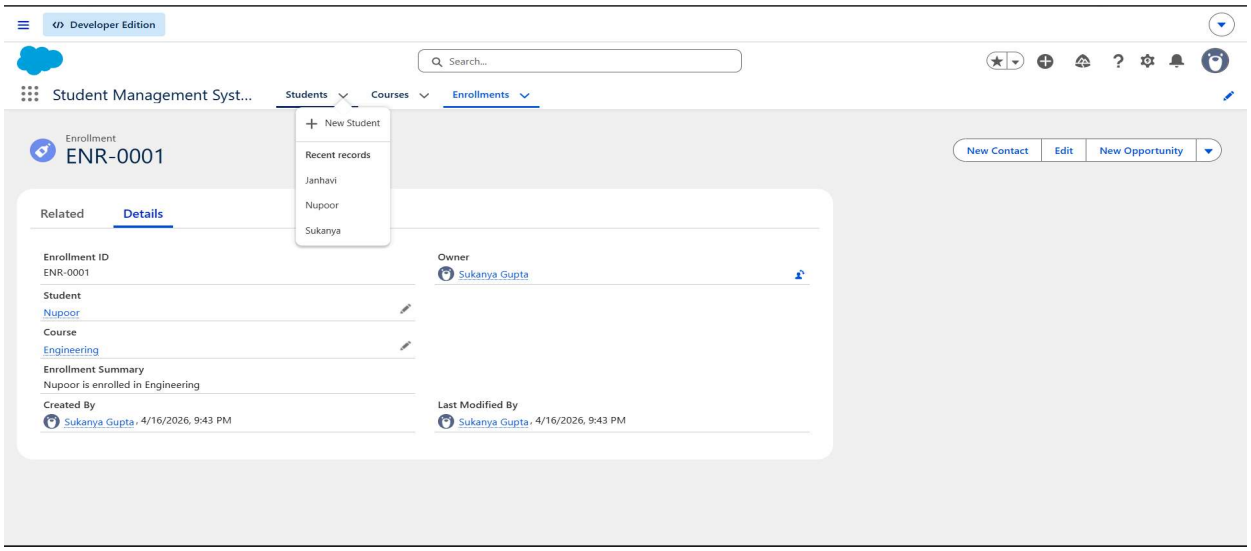
Step 9: We opened the Student Management System application in Lightning Experience. The navigation bar shows three tabs: Students, Courses, and Enrollments. Under the Courses tab, we created a new Course record named 'Engineering', which appears in the Recently Viewed list, confirming the custom object and app navigation are working correctly.



Step 10: We created an Enrollment record ENR-0001, linking Student 'Nupoor' to Course 'Engineering'. The record detail page displays the Enrollment ID (ENR-0001), Student (Nupoor), Course (Engineering), and the automatically computed Enrollment Summary formula field showing: 'Nupoor is enrolled in Engineering' - confirming the formula field is working correctly.



Step 11: We verified the Students data by clicking the Students dropdown in the navigation bar, which revealed the recently added student records: Janhavi, Nupoor, and Sukanya. The enrollment record ENR-0001 details remain visible in the background, showing the complete working state of the Student Management System with multiple students enrolled.



Conclusion

In this experiment, we successfully designed and developed a Student Management System as a custom mini project application on the Salesforce Cloud platform. We created three custom objects Student_c, Course_c, and Enrollment_c with appropriate Lookup Relationship fields linking them together. We added an Auto Number field for automatic Enrollment ID generation and a Formula Field (Enrollment Summary) that dynamically generates text combining the student and course names. A Lightning Experience app was created using the App Manager, exposing all three objects as navigable tabs. The application was tested by creating student records (Janhavi, Nupoor, Sukanya), a course record (Engineering), and an enrollment record (ENR-0001), with the formula field correctly displaying 'Nupoor is enrolled in Engineering'. This demonstrated how Salesforce's declarative tools enable rapid development of relational cloud applications without writing any code.